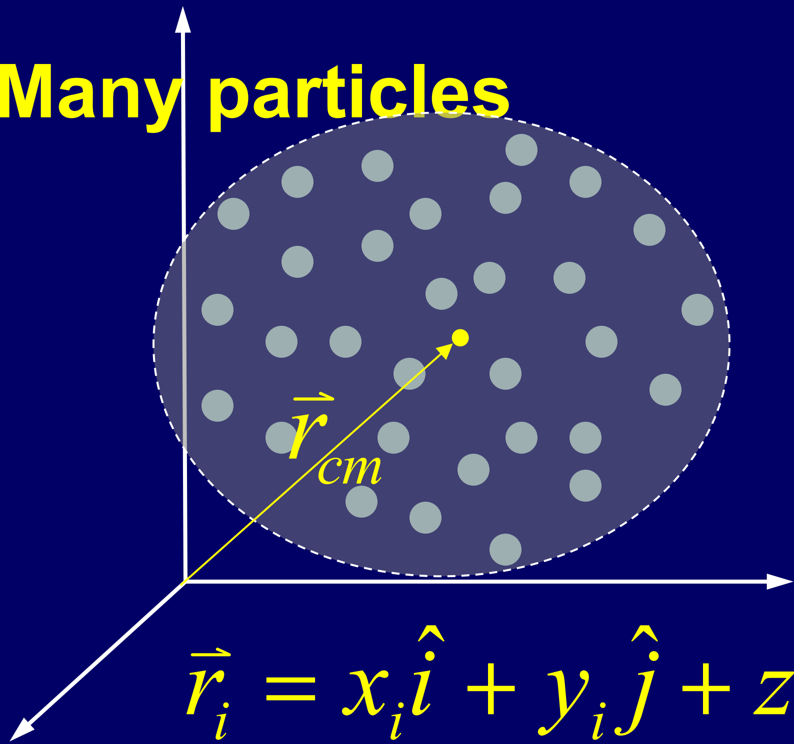
The center of mass

Many particles



$$\vec{r}_{cm} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + \cdots + m_N\vec{r}_N}{m_1 + m_2 + \cdots m_N} = \frac{1}{M}\sum m_i\vec{r}_i = \frac{1}{M}\int \vec{r}\,dm$$

$$\vec{r}_i = x_i\hat{i} + y_i\hat{j} + z_i\hat{k}$$

$$x_{cm} = \frac{1}{M}\sum m_i x_i = \frac{1}{M}\int x\,dm,$$

$$M = m_1 + m_2 + \cdots + m_N = \sum m_n.$$

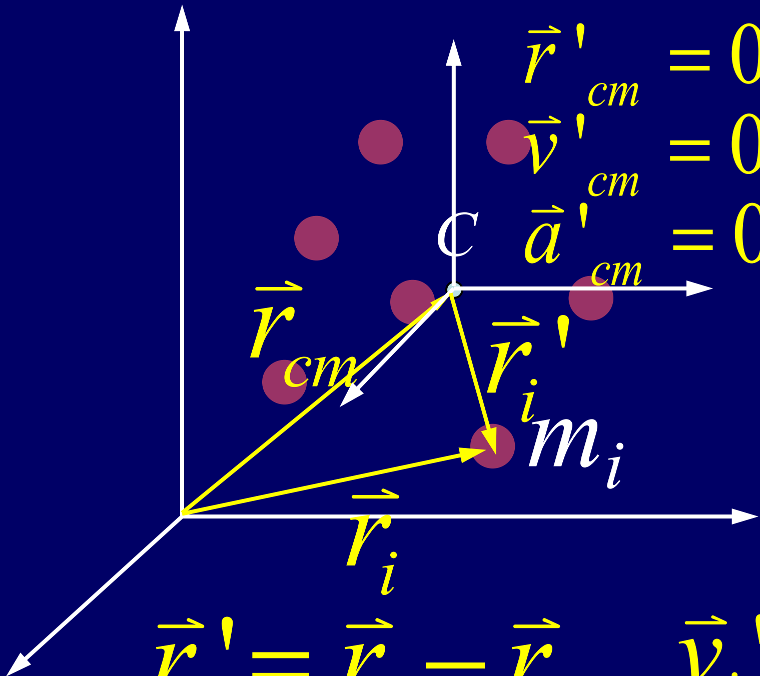
$$y_{cm} = \frac{1}{M}\sum m_i y_i = \frac{1}{M}\int y\,dm,$$

$$\vec{v}_{cm} = \frac{1}{M}\int v\,dm \qquad \vec{a}_{cm} = \frac{1}{M}\int a\,dm$$

$$z_{cm} = \frac{1}{M}\sum m_i z_i = \frac{1}{M}\int z\,dm.$$

$$M\vec{v}_{cm} = \vec{P} \qquad M\vec{a}_{cm} = \sum \vec{F}_{ext}$$

Lab Frame **cm Frame**



$$\vec{r}\,'_{cm} = 0$$
$$\vec{v}\,'_{cm} = 0$$
$$\vec{a}\,'_{cm} = 0$$

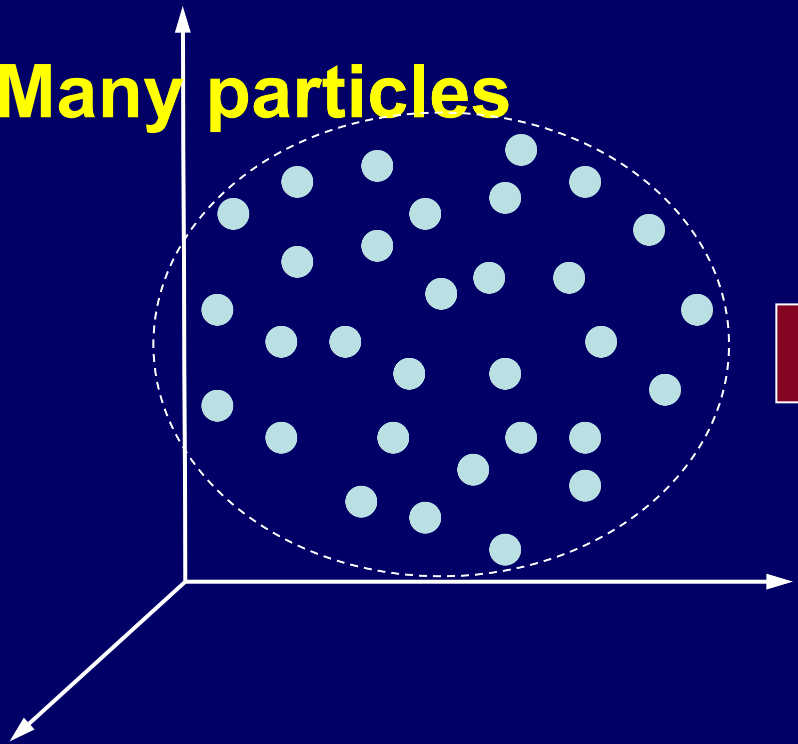
In the **cm frame** the total momentum is zero

The **cm frame** called zero momentum reference frame.

$$\vec{r}_i\,' = \vec{r}_i - \vec{r}_{cm} \qquad \vec{v}_i\,' = \vec{v}_i - \vec{v}_{cm} \qquad \vec{a}_i\,' = \vec{a}_i - \vec{a}_{cm}$$

$$\vec{P}\,' = \sum^{N} m_i \vec{v}_i\,' = M\vec{v}\,'_{cm} = 0$$

Many particles



One particle

$$\Sigma\vec{F} = m\vec{a}$$

$$\vec{p} = m\vec{v}$$

$$\Sigma\vec{F} = \frac{d\vec{p}}{dt}$$

Particle system

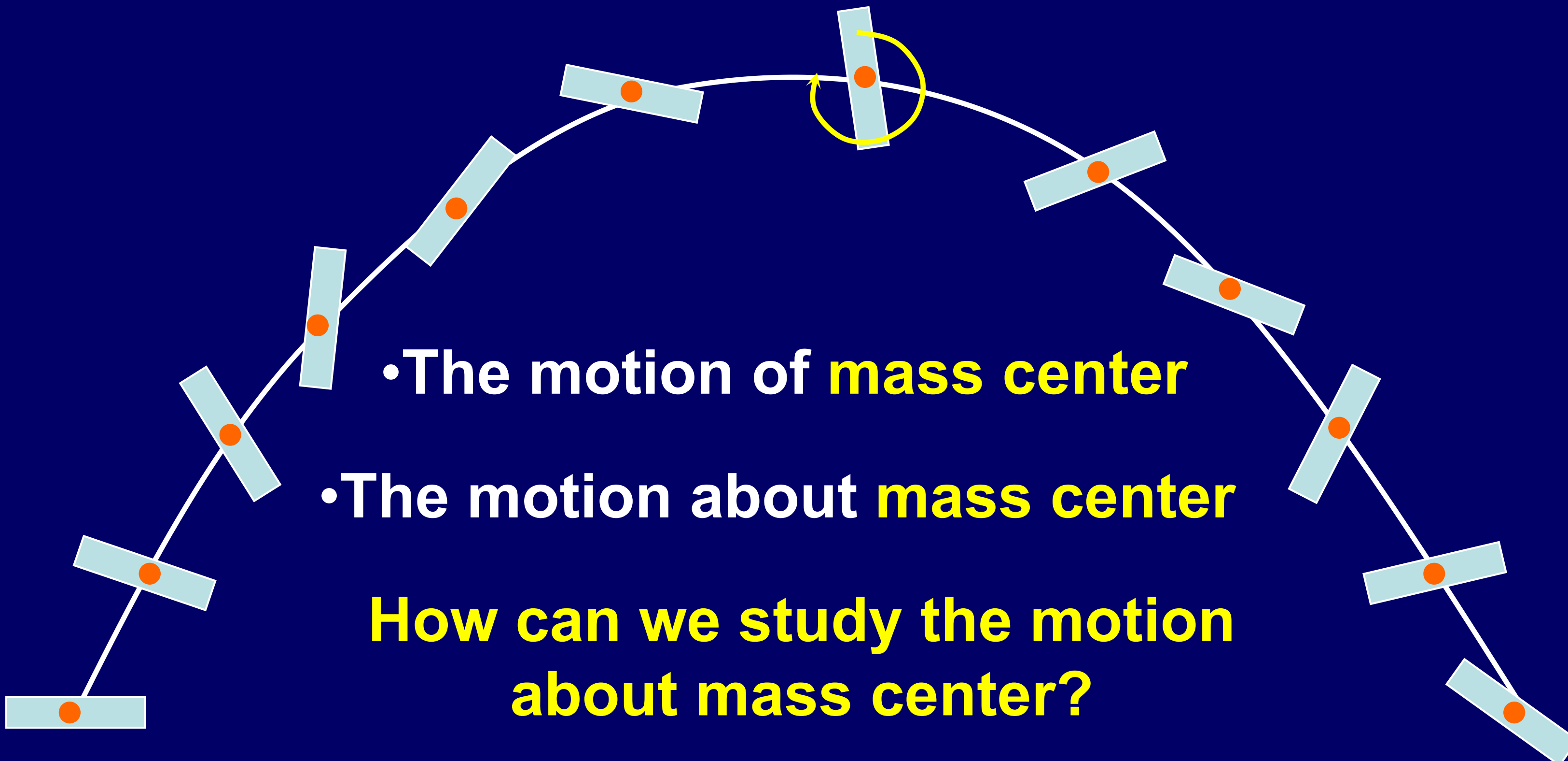
$$\Sigma\vec{F}_{ext} = M\vec{a}_{cm}$$

$$\vec{P} = M\vec{v}_{cm}$$

$$\Sigma\vec{F}_{ext} = \frac{d\vec{P}}{dt}$$

Motion of solid body

$$\sum \vec{F}_{ext} = M\vec{a}_{cm}$$



- The motion of **mass center**
- The motion about **mass center**

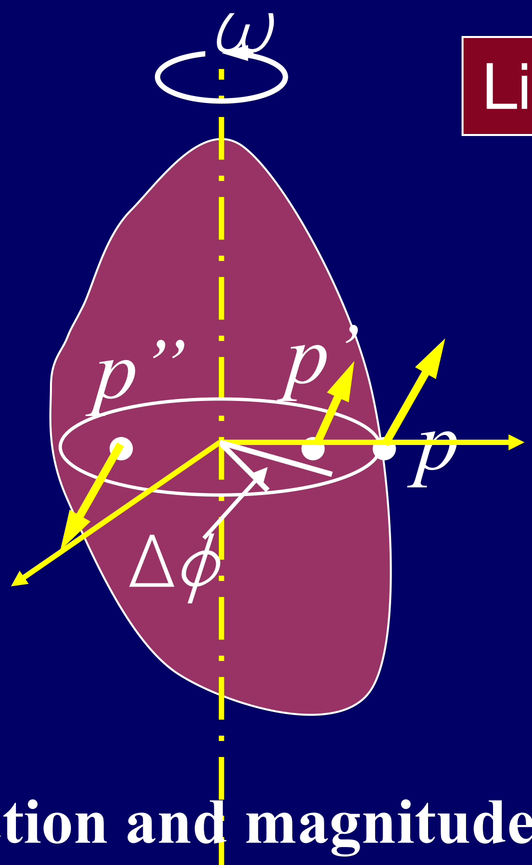
How can we study the motion about mass center?

Rotational motion

Chapter 8

Rotational Kinematics

Rotational motion



Linear motion	Rotational motion
ΔS	$\Rightarrow \Delta\varphi$
v	$\Rightarrow \omega$
a	$\Rightarrow \alpha$

average angular acceleration

$$\alpha_{av} = \frac{\Delta \omega}{\Delta t}$$

Direction and magnitude of the velocity is different

$$\alpha = \lim_{\Delta t \to 0} \frac{\Delta\omega}{\Delta t} = \frac{d\omega}{dt}$$

The rotational motion is the same for *p* and *p'*, *p''*

instantaneous angular acceleration

Linear motion | **Rotational motion**

$$v = \frac{dr}{dt} \quad \Rightarrow \quad \omega = \frac{d\varphi}{dt}$$

$$a = \frac{dv}{dt} \quad \Rightarrow \quad \alpha = \frac{d\omega}{dt}$$

Rotation with constant angular acceleration

$$d\omega = \alpha \, dt$$

$$v = v_0 + at \qquad \omega = \omega_0 + \alpha t$$

$$s = s_0 + v_0 t + \frac{1}{2} a t^2 \qquad \varphi = \varphi_0 + \omega_0 t + \frac{1}{2} \alpha t^2$$

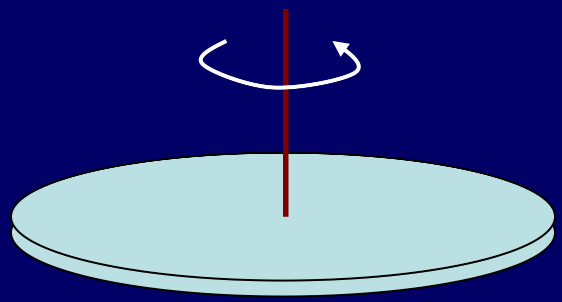
$$v^2 = v_0^2 + 2a(s - s_0) \qquad \omega^2 = \omega_0^2 + 2\alpha(\varphi - \varphi_0)$$

Example

$$\alpha = const.\ \ t = 0,\ \ \omega_0 = 20\, rad/s$$

$$After\ 40rev,\ \ \omega = 0\, rad/s$$

$$How\ long\ time\ does\ it\ stop\,?$$



$$\phi - \phi_0 = 40rev = 80\pi$$

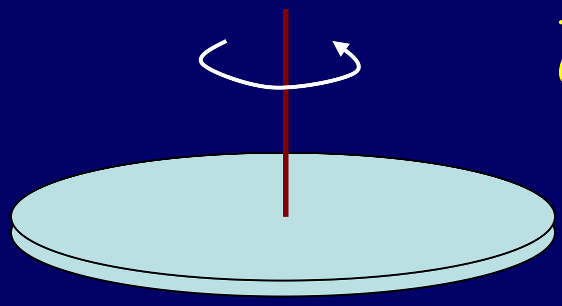
$$\omega^2 = \omega_0^2 + 2\alpha(\phi - \phi_0) \qquad \alpha = \frac{\omega^2 - \omega_0^2}{2(\varphi - \varphi_0)}$$

$$\omega = \omega_0 + \alpha t \qquad t = \frac{\omega - \omega_0}{\alpha} = \frac{-20}{-\frac{5}{2\pi}} = 8\pi(s)$$

Example

$$\phi = 2t - 3t^2 + t^3 \quad \omega\big|_{t=1} = ? \; \omega\big|_{t=2} = ?$$

$$\overline{a}\big|_{t=1,t=2} = ? \quad a\big|_{t=1} = ? \quad a\big|_{t=2} = ?$$



$$\omega = \frac{d\varphi}{dt} = 2 - 6t + 3t^2$$

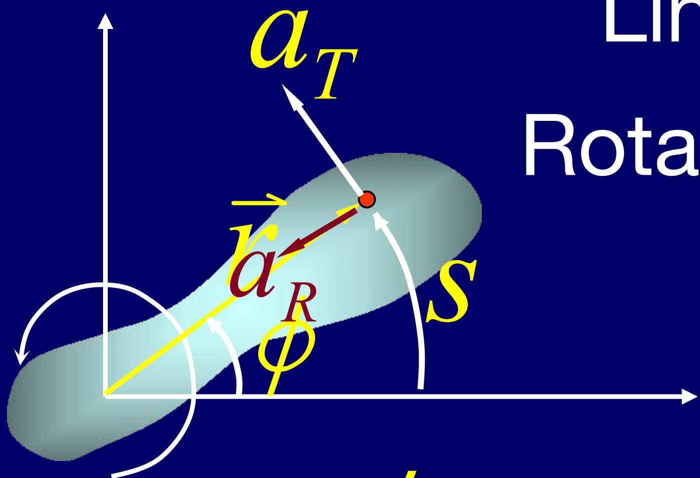
$$\omega\big|_{t=1} = -1, \; \omega\big|_{t=2} = 2$$

$$a = \frac{d\omega}{dt} = -6 + 6t$$

$$\overline{a} = \frac{\omega_2 - \omega_1}{2-1} = 3$$

$$a\big|_{t=1} = 0, \quad a\big|_{t=2} = 6$$

Relationships between linear and angular variables



Linear motion	s	v	a
Rotational motion	ϕ	ω	α

$$v = \frac{ds}{dt} = \frac{d(\phi r)}{dt} = \frac{d\phi}{dt} r = \omega r$$

$$s = \phi r$$

$$a_T = \frac{dv}{dt} = \frac{d\omega}{dt} r = \alpha r$$

$$\vec{v} = \vec{\omega} \times \vec{r}$$

$$\vec{a}_T = \vec{\alpha} \times \vec{r}$$

$$a_R = \frac{v^2}{r} = \frac{(\omega r)^2}{r} = \omega^2 r$$

$$\vec{a}_R = -\omega^2 \vec{r}$$

$$\vec{a} = \vec{a}_T + \vec{a}_R$$

Rotational quantities as vectors

$$v = \omega r$$

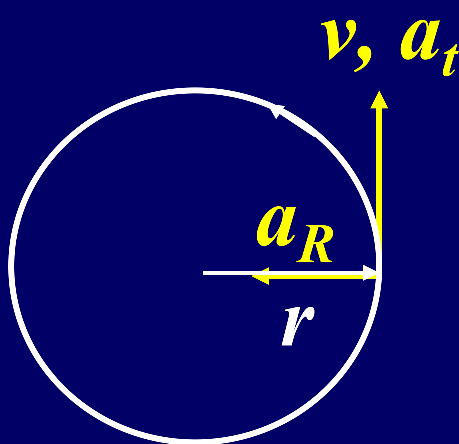
$$\vec{v} = \vec{\omega} \times \vec{r}$$

$$\vec{a} = \frac{d(\vec{\omega} \times \vec{r})}{dt}$$

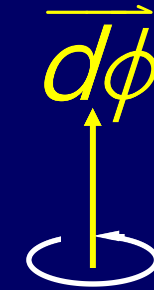
$$= \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt}$$

$$= \vec{\alpha} \times \vec{r} + \vec{\omega} \times \vec{v}$$

$$= \vec{a}_T + \vec{a}_R$$



Angular displacement $\overrightarrow{d\phi}$



Angular velocity $\vec{\omega} = \dfrac{\overrightarrow{d\phi}}{dt}$

Angular acceleration $\vec{\alpha} = \dfrac{d\vec{\omega}}{dt}$

$$\vec{a}_R = \vec{\omega} \times \vec{v} = \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

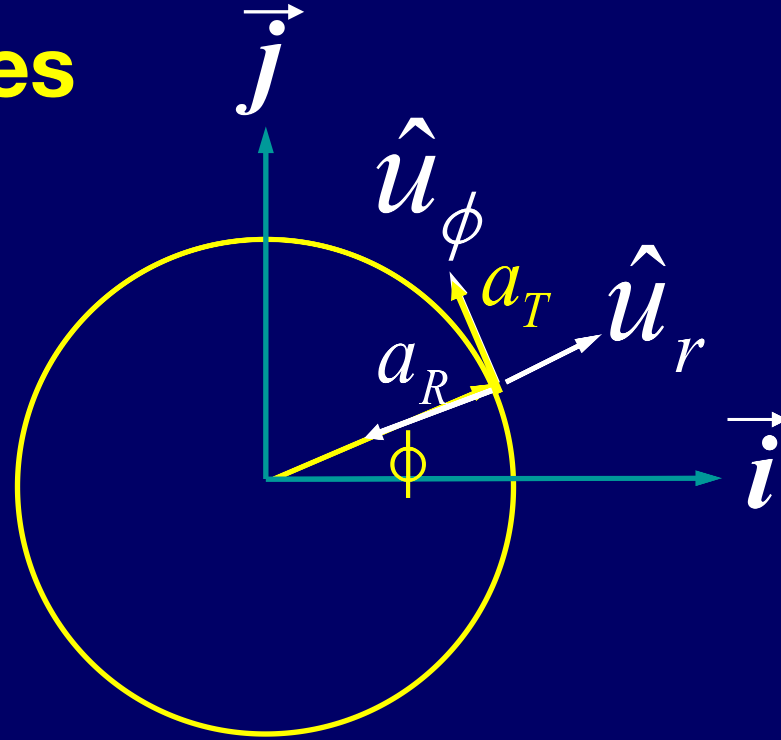
$$= (\vec{\omega} \cdot \vec{r})\vec{\omega} - (\vec{\omega} \cdot \vec{\omega})\vec{r} = -\omega^2 \vec{r}$$

In plane polar coordinates

$$\hat{u}_r = \cos\phi\,\hat{i} + \sin\phi\,\hat{j}$$

$$\hat{u}_\iota = -\sin\phi\,\hat{i} + \cos\phi\,\hat{j}$$

$$\frac{d\hat{u}_\phi}{dt} = -\omega\hat{u}_r \qquad \frac{d\hat{u}_r}{dt} = \omega\hat{u}_\phi$$



$$\vec{v} = v\hat{u}_\phi$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d(v\hat{u}_\phi)}{dt} = \frac{dv}{dt}\hat{u}_\phi + v\frac{d\hat{u}_\phi}{dt} = \frac{dv}{dt}\hat{u}_\phi - v\omega\hat{u}_r = \frac{dv}{dt}\hat{u}_\phi - \frac{v^2}{r}\hat{u}_r$$

$$= a_T\hat{u}_\phi - a_R\hat{u}_r$$

$$\Delta r,\ v,\ a \ \text{ are vectors}$$

$$\Delta \vec{r},\ \vec{v},\ \vec{a}$$

$$\text{Are}\ \Delta\varphi,\ \omega,\ \alpha \ \text{ vectors?}$$

$$\omega,\ \alpha \ \text{ are vectors}$$

$$\Delta\vec{\varphi}\,?$$

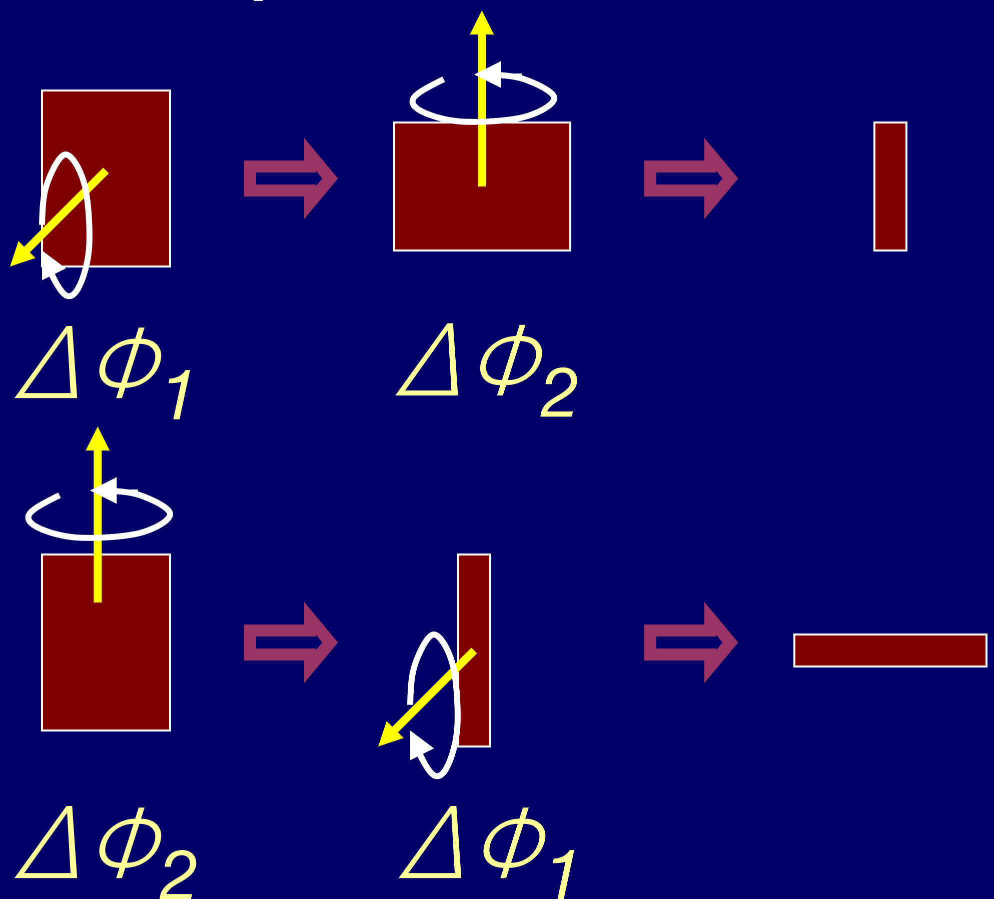
$$\vec{\omega}, \vec{\alpha}$$

$$\vec{v} = \vec{\omega} \times \vec{r}$$

$$\vec{a}_T = \vec{\alpha} \times \vec{r}$$

$$\vec{a}_R = \vec{\omega} \times \vec{v}$$

Rotational quantities as vectors

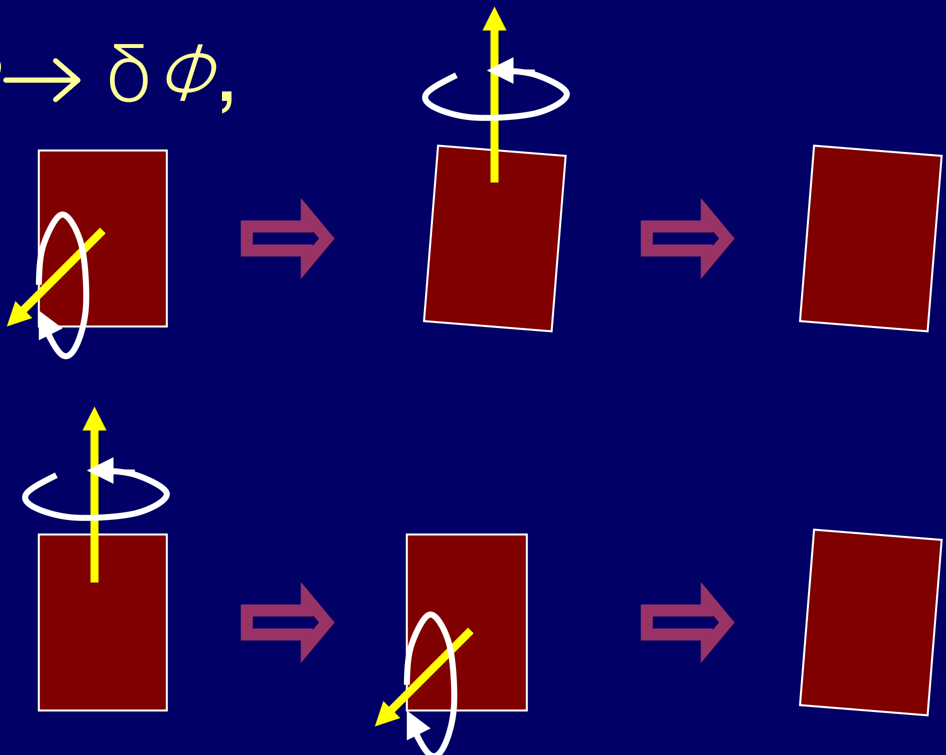


$\Delta\Phi_1 + \Delta\Phi_2 \neq \Delta\Phi_2 + \Delta\Phi_1$

$\Delta\Phi$ is not a vector

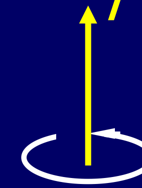
Rotational quantities as vectors

$\Delta\Phi \to \delta\Phi$,



$\overrightarrow{d\phi}$

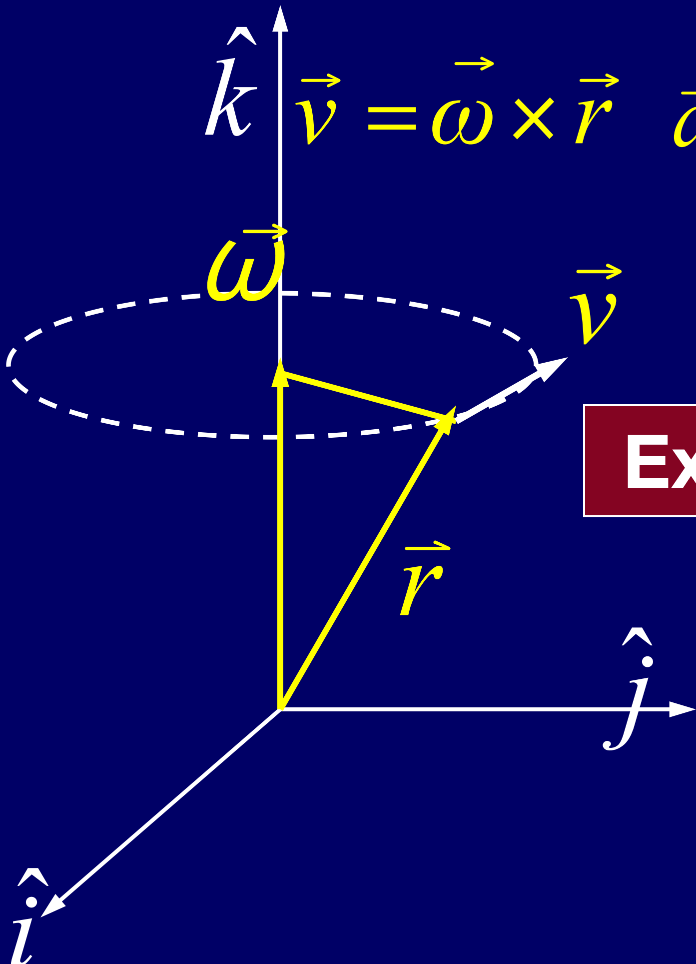
$\overrightarrow{d\phi}$



$$\vec{\omega} = \frac{\overrightarrow{d\phi}}{dt}$$

$$\vec{\alpha} = \frac{d\vec{\omega}}{dt}$$

$\delta\Phi_1+\delta\Phi_2=\delta\Phi_2+\delta\Phi_1$ $\delta\Phi$ is a vector



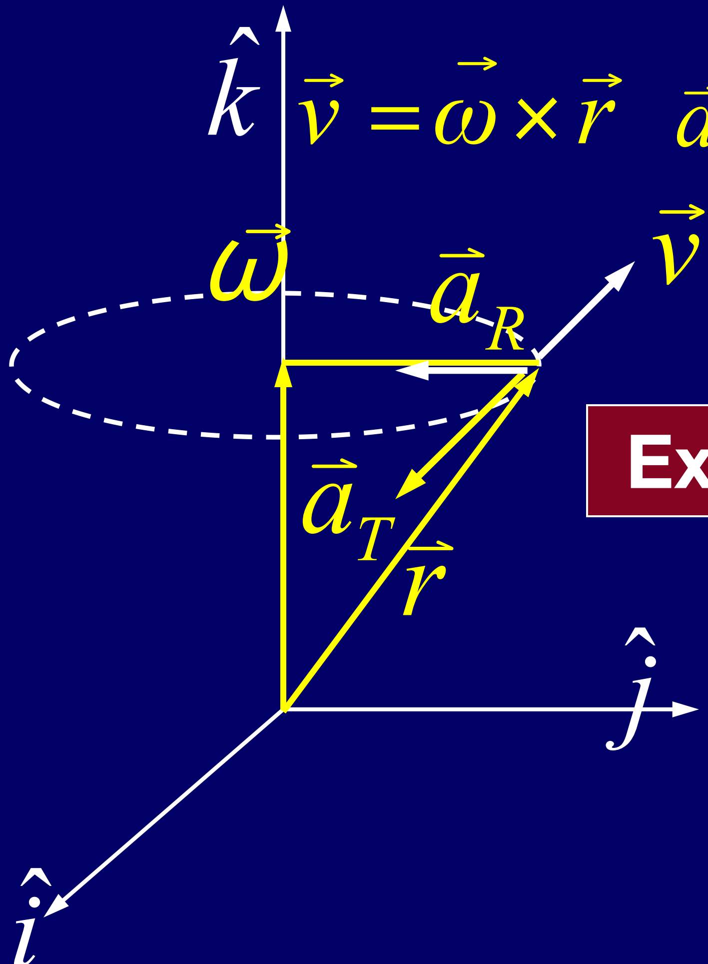
$$\vec{v} = \vec{\omega} \times \vec{r}$$

$$\vec{a} = \frac{d(\vec{\omega} \times \vec{r})}{dt} = \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt}$$

$$= \vec{\alpha} \times \vec{r} + \vec{\omega} \times \vec{v} = \vec{a}_T + \vec{a}_R$$

Example

$A \ particle \ \vec{r} = 2\hat{j} + 3\hat{k}$



$$\vec{v} = \vec{\omega} \times \vec{r}$$

$$\vec{a} = \frac{d(\vec{\omega} \times \vec{r})}{dt} = \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt}$$

$$= \vec{\alpha} \times \vec{r} + \vec{\omega} \times \vec{v} = \vec{a}_T + \vec{a}_R$$

Example

$$A \ particle \ \ \vec{r} = 2\hat{j} + 3\hat{k}$$

$$\vec{\omega} = 4\hat{k} \quad \vec{\alpha} = -2\hat{k}$$

$$\vec{v} = \vec{\omega} \times \vec{r} = -8\hat{i}$$

$$\vec{a} = \vec{a}_T + \vec{a}_R = 4\hat{i} - 32\hat{j}$$

$$\vec{a}_T = \vec{\alpha} \times \vec{r} = 4\hat{i} \quad \vec{a}_R = \vec{\omega} \times \vec{v} = -32\hat{j}$$

Force and Motion

The force is an interaction between a body and its environment.

The force acting on a body will accelerate it

$$\vec{F} = m\vec{a}$$

How about the Force and Rotation

CHAP 8

Exercises

P171: 3,30,32

Problems

P173: 6, 7